THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The average of 48 numbers is 80. The average of all 105 numbers (including 57 more numbers) is 61. The average of the 57 additional numbers is: 1

(a) 46 **(b)** 47 **(c)** 44 **(d)** 45

2. A shop earns a profit of 15% on sales of ₹2000 and 45% on sales of ₹3000. The overall profit percentage on total sales is: 1

(a) 33% **(b)** 35% **(c)** 34% **(d)** 32%

3. 910 mL of a 18% sugar solution is mixed with 780 mL of a 5% sugar solution. The concentration of the mixture is: 1

(a) 13% **(b)** 14% **(c)** 11% **(d)** 12%

4. The average marks of 50 students is 94 and of another 150 students is 66. The average of all 200 students is: 1

(a) 72 **(b)** 75 **(c)** 74 **(d)** 73

5. A group of 20 people has average weight 81 kg. Another group has average weight 48 kg. Together the average weight is 53 kg. The number of people in the second group is: 1

(a) 113 **(b)** 111 **(c)** 112 **(d)** 114

6. In what ratio must a 54% solution and a 89% solution be mixed to obtain a 72% solution? 1

(a) $17 : 19$ **(b)** $18 : 18$ **(c)** $17 : 18$ **(d)** $18 : 17$

7. A bus goes from A to B at 28 km/h and returns along the same road at 70 km/h. The average speed for the whole journey (km/h) is: 1

(a) 40 **(b)** 39 **(c)** 42 **(d)** 41

8. The average monthly salary of 19 employees is ₹52 thousand. An employee earning ₹24 thousand leaves and a new employee joins, and the average rises by ₹4 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 105 **(b)** 96 **(c)** 100 **(d)** 104

9. **Assertion (A):** In a 100% stacked bar chart every bar has the same total length. 1
Reason (R): In a 100% stacked bar chart each bar shows the shares of its parts as percentages, which add up to 100%.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** If 2 kg of tea costing Rs 300 per kg is mixed with 3 kg of tea costing Rs 400 per kg, the mixture costs Rs 360 per kg. 1

Reason (R): The cost per kg of the mixture is the total cost divided by the total mass.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The average monthly salary of 26 employees is ₹74 thousand. An employee earning ₹20 thousand leaves and a new employee joins; the average rises by ₹5 thousand. Find the new employee's monthly salary. 2

12. A chemist has a 6% acid solution and a 51% acid solution. In what ratio must they be mixed to get a 32% solution? 2

13. A cyclist rides from home to a market at 36 km/h and returns by the same route at 45 km/h. Find the average speed for the whole trip. 2

14. The average monthly salary of 22 employees is ₹56 thousand. An employee earning ₹36 thousand leaves and a new employee joins; the average rises by ₹4 thousand. Find the new employee's monthly salary. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a class, 63 students have an average of 92 marks and the remaining students have an average of 81 marks. The class average is 88. Find the number of remaining students. 3

16. A final score combines Maths (score 96, weight 5), Science (score 52, weight 7), English (score 74, weight 3), Hindi (score 66, weight 4). Find the weighted average score. 3

17. The mean of 10 numbers is 57. Two more numbers 70 and 56 are added. Find the new mean and say whether it rises or falls. 3

OR

In a class, 27 students have an average of 54 marks and the remaining students have an average of 87 marks. The class average is 78. Find the number of remaining students. 3

18. A final score combines Maths (score 90, weight 6), Science (score 67, weight 2), English (score 56, weight 3), Hindi (score 58, weight 6). Find the weighted average score. 3

OR

The mean of 18 numbers is 31. Two more numbers 55 and 27 are added. Find the new mean and say whether it rises or falls. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Anil's final score in a course is a weighted average. His Test score is 85 with weight 2, his Project score is 95 with weight 3 and his Exam score is 75 with weight 5. 5

(i) Find the sum of the weights.

(ii) Find the sum of weight × score.

(iii) Find Anil's weighted average.

(iv) Find the simple mean of the three scores and explain why the weighted average is lower.

20. A trader mixes rice costing ₹65 per kg and rice costing ₹185 per kg in the ratio 3:1. Find the cost price per kg of the mixture and the selling price per kg to earn a profit of 20%. 5

OR

A family spent ₹3700 in January and ₹5900 in February. Food took 55% of the January total and 25% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks